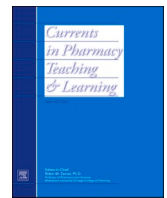




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Development and evaluation of AI chatbot tool for written communication training in self-care: Experiences of pharmacy students and faculty

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ABSTRACT

Background: Effective communication is crucial in pharmacy practice, particularly in self-care counseling. As online pharmacies and chat-based consultations expand, training in digital written communication is increasingly important. Artificial intelligence (AI) systems based on large language models (LLMs) offer a structured and engaging environment to support skill development through conversational agents. This study explored the use of LLM-based chatbots to train pharmacy students in written synchronous communication for self-care consultations.

Methods: Three chatbot-simulated patients and an LLM-based feedback system were developed to reflect common self-care scenarios and provide communication-focused feedback. Fourteen pharmacy students and faculty interacted with the chatbots and shared their experiences through semi-structured interviews. Thematic analysis was used to identify patterns in the data.

Results: The analysis identified five main themes. Participants emphasized the authenticity of the simulated patient interactions, particularly their emotional realism. The AI-generated feedback was described as structured, detailed, and fair especially valued for its focus on communication skills. Faculty appreciated the consistency of the feedback and highlighted its added value to complement human assessment. Students discussed the cognitive and emotional demands of the experience, suggesting potential to tailor chatbot complexity to learners' needs.

Conclusion: LLM-based chatbots represent a pedagogically grounded and scalable tool for developing pharmacy students' written communication skills in self-care consultations. This approach offers a foundation for building shared virtual patient infrastructures and integrating communication theory into digital education. It holds promise for broad implementation across pharmacy programs adapting to the demands of online and hybrid care.

Background

Effective communication skills are essential for pharmacists to improve patient health outcomes.^{1,2} Yet, in everyday pharmacy practice, pharmacists do not consistently use communication strategies that are known to improve patient outcomes.^{3,4} Various teaching methods exist to develop these skills, ranging from more traditional lectures to more effective and active approaches such as

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role-play and simulated patient interactions.^{5,6} Nordic pharmacy students have expressed a need for more active teaching methods, particularly the ones capable of providing individualized feedback.⁷ However, active teaching methods are resource-intensive and often make it challenging to provide consistent and equitable feedback to all students.^{8,9}

Artificial intelligence (AI), particularly large language models (LLMs), has been proposed as powerful tools for not only analyzing but also teaching communication skills.¹⁰ AI has the potential to offer low-stakes, repeated practice, tailored feedback, and more responsive simulated patients, in contrast to traditional digital simulated patient systems, which operate based on predefined dialogue trees and offer limited interactivity in response to learners' actions.^{9–12} In 2023 Stamer's review identified 12 studies of health communication education, highlighting three key trends in how AI has been used so far: the integration of AI in virtual realities; the use of natural language processing for text analysis and information extraction; and for simulation of virtual patients.¹² Examples of these trends include virtual reality for interprofessional team training in nursing,¹³ the ReadMI platform in the U.S., which uses AI to provide real-time feedback on motivational interviewing performance,¹⁴ and the ATLAS platform for nutrition and dietetics education, where students interact with LLM-based simulated patients and receive automated feedback.¹⁵ Although evidence on learning outcomes is still limited, these tools could potentially support the development of communication skills, and offer more consistent exposure to key communicative situations.^{9,12}

Despite the rapid development of AI in health profession education, a recent review by Abdel et al. identified only seven published papers on AI in pharmacy education,¹⁶ with just one focusing on improving communication skills.¹⁷ Further has the rise of online pharmacies globally increased the need to train pharmacists also in written communication skills.^{18,19} As self-care medicines are increasingly purchased online, digital two-way communication—especially synchronous formats like chat¹⁹—is emerging as a natural extension of pharmacy services, with growing expectations that pharmacists will be able to support patients through these channels. Chatbots could offer a valuable tool in this training, enabling students to practice real-time written communication with simulated patients, receive automated feedback, and build confidence in a low-stakes environment. Developing training activities focused on written communication is therefore both timely and pedagogically relevant. Given the combination of advancing technology and evolving pharmacy practice, this study explores pharmacy students' and faculty's reactions to and the perceived feasibility of using LLM-based chatbots to train written synchronous communication skills in self-care consultations.

Educational activity- method

Study design

The study consisted of two main phases: the development of ChatGPT-based models (simulated patients and automated feedback) and the evaluation of their feasibility through participant testing and interviews. In the first phase, ChatGPT²⁰ (GPT-4o) was used to create interactive learning models. A multi-step process was utilized to design simulated patients with whom students could engage in synchronous chat to practice communication skills (Fig. 1). Three simulated patients were developed to replicate real-world self-care scenarios, incorporating diverse symptoms and emotional states (Table 1). The automated feedback system focused on key communication skills relevant to self-care consultations, which form a core part of the communication training provided to our pharmacy students (Appendix A Fig. A1.A). The feedback system was based on a consultation framework commonly used in pharmacy education and in pharmacies across Sweden, known as “The communication ladder” (Appendix B). This framework consists of four phases designed to map a patient's self-care needs. Structured feedback was provided on how well the consultation framework's phases was applied. The feedback system targeted core communication skills emphasized in the curriculum²¹, such as open and closed questioning, summarizing, paraphrasing, and empathic responding, by identifying which skills were used (e.g., through frequency counts), providing positive examples, and suggesting areas for improvement. It also evaluated whether students asked relevant, content-specific questions to explore the patient's needs. A final summary synthesized the feedback (Appendix C). The optimization of

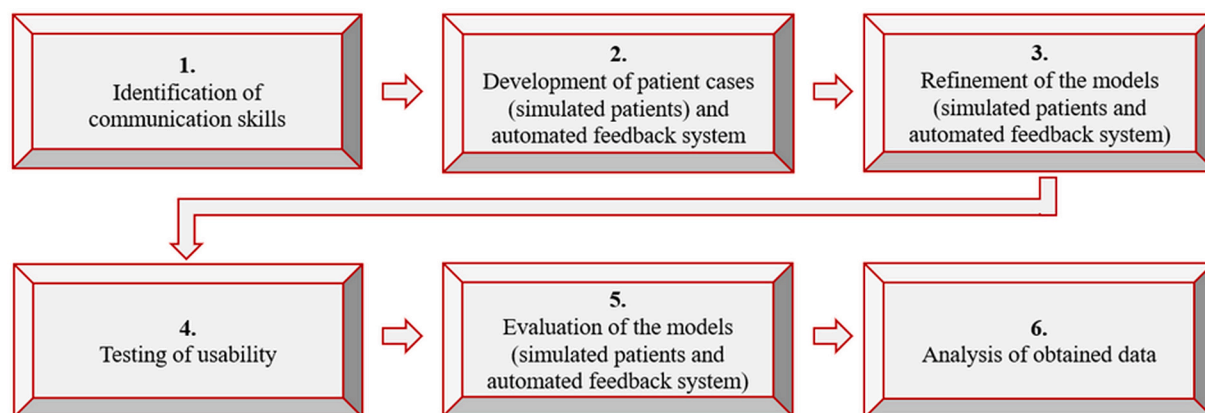


Fig. 1. Development process of the ChatGPT-based communication training.

Table 1

Overview of the three simulated patients developed to simulate real-world self-care scenarios, incorporating a variety of symptoms and emotional states.

	Patient 1 – Omar 65 years old	Patient 2 – Maria 30 years old	Patient 3 – Natalie 50 years old
General	65-year-old with a cold - Symptoms: Dry cough, nasal congestion, runny nose, sore throat - Has had the symptoms for three days - Has no other medical conditions - Takes no other medications - Has not tried any over the counter medications for the symptoms	- Wants treatment for pinworms - Symptoms: Itching around the anus during the evening - For her 4-year-old son - Has seen the worms, they look like small white larvae - Symptoms have been present for one day - No other family members have symptoms - Has not had worms before - No other medical conditions - Takes no other medications - There are three people in the family - Maria (the mother) is pregnant - Has not tried any over the counter medications for the symptoms - Afraid	50-year-old woman needs help to quit smoking - Has a brother who was a smoker and recently passed away from lung cancer - Has never attempted to quit smoking before - Has been smoking for 20 years - Smokes about one pack (20 cigarettes) per day - Often smokes together with her partner - Has asthma and high blood pressure - Takes ventoline diskus inhalation powder for asthma and propranolol for high blood pressure - Has not tried any nicotine replacement therapies before
Emotional state	- Tired - Stressed		- Shy - Sad - Insecure
Extra	He believes he already has the cough medicine at home that the pharmacist recommended, and asks if he can return the medicine tomorrow if that's the case	- Her son cannot swallow tablets - Wonders what more she can do in addition to giving medication	- Has had some headaches recently due to everything surrounding her brother's passing. - Asks where to find Ipren® (ibuprofen) and whether it can be used for headaches.

process involved setting up clear instructions, rules, and contextual parameters to ensure that the simulated patient interactions and feedback system produced realistic, relevant, and pedagogically useful responses, see Appendix A Fig. A1.B and Appendix D (Final prompt).²² For example, if a an “angry” patient were too aggressive the system was given instructions to balance this. Before testing with students, a comparison was made between the AI-generated feedback and faculty-provided feedback on a sample of six conversations. The two were judged to be of comparable quality (Appendix E) by SB and KS. In total, the system was refined in more than 100 iterative rounds across approximately four weeks. In the second phase, semi-structured interviews were conducted with all participants immediately after completing the simulated patient consultations, see below.

The research team

The research team comprised individuals from diverse professional backgrounds. SB was a Master's student in pharmacy (female). KS and EL are pharmacists (both PhD, females) with extensive experience in qualitative research. MS is a MD (male, PhD) with broad expertise in medical education, and KY (male, PhD) has a background in computer science, artificial intelligence, and education research. SB had no prior relationship with the students, but had met some of the faculty. KS and EL, as part of the faculty, are colleagues of those who participated in the study.

Study population and recruitment

Participants were selected through purposive sampling from two mandatory self-care courses in the Pharmacy program at Uppsala University. To ensure diverse perspectives, both students and faculty from these courses were included. Recruitment of students was conducted via Zoom (Zoom Video Communications Inc., Version 6.2. 2024) presentations or in-person during class lectures. Faculty were contacted separately via email, with detailed information about the study objectives and participation requirements.

Data collection

Data collection occurred during October and November 2024. SB carried out the interviews after training and supervision from EL and KS. First, participants engaged in a virtual patient self-care consultation. Students logged in to ChatGPT using a laptop and a paid subscription account provided by us, where they accessed a page with the different patients. The patient described in Appendix D can be accessed through this custom GPT: <https://chatgpt.com/g/g-68ac518a81048191bb0f053e72d03048-chatbot-for-communication-training-in-self-care>. Each test round lasted approximately ten minutes (chatting with one patient), after which the models generated

real-time feedback on participants' communication skills (Appendix C) in the same chat. Participants then reviewed and reflected on the feedback prior to moving on to the second phase. In the second phase, participants underwent one-on-one semi-structured interviews with researchers either in-person at the University or via Zoom. The semi-structured interview guides used in this study (Appendix F) were validated through field testing by individuals from the same target group.²³ All interviews were recorded and transcribed verbatim for subsequent analysis.

Data analysis

The study utilized Level 1 (Reaction) of Kirkpatrick's Four-Level Training Evaluation Model to assess participants' immediate experiences and the practicality of the chatbot tool.²⁴ An inductive descriptive content analysis was employed.²⁵ Data were structured in Microsoft Word for Windows, version 2021. The first step involved a review of all transcripts to gain familiarity with the data. Each transcript was then analyzed separately by SB, and responses were summarized. Coding followed Tjora's guidelines for creating effective empirical codes.²⁶ Codes were then reviewed by EL and KS. All codes were grouped by topic. Topics were further classified into subthemes and broader themes and discussed among the researchers.²⁵ Each theme was then reviewed to ensure coherence and relevance to the research question. In the last step analytic text was written.

Ethical considerations

This study did not require formal ethical approval under the Swedish Ethical Review Act, as did not involve collection of sensitive personal data.²⁷ Nonetheless, participants were informed about the voluntary nature of the study. Written informed consent was obtained from all participants, and interview responses were completely de-identified for privacy and confidentiality. Access to data was limited to researchers directly involved in data collection and analysis.

Evaluation findings

In total, 14 participants took part in the study, including five faculty and nine students (12 female, two male). Gender was self-reported and recorded during demographic data collection. The interviews lasted approximately 25 min on average (11–55 min). The analysis identified five main themes and 12 subthemes, highlighting various aspects of the simulated patient interaction and automated feedback, regarding their perceived usefulness in education (Table 2).

Simulated but meaningful: Bridging simulation and real-world

Faculty and students found the simulated patients easy to use, which supported initiating and conducting conversations. The interactions were described as natural, similar to everyday chat-platforms. However, participants expressed a need for clearer instructions to better understand the tool's available functions, such as attaching files, generating images, and editing messages. All participants saw the simulated patients and automated feedback system as a valuable complement to existing training. Students appreciated the opportunity for additional communication practice, while faculty highlighted the benefit of extra training in a format resembling examination. Both groups valued the suitability of the simulated patients for individual use when time allowed, as well as the ability to review past conversations and feedback to support reflection and improvement:

"When you're practicing on your own and don't have someone else [another student] to role-play with, it's quite difficult to practice communication. In contrast, here you have someone [the patient] that is more interactive, and the responses are adapted to what you say." (Student 4).

The chatbot tool was also considered useful in group work, where students could compare their responses and discuss different ways of handling specific patient scenarios. Some participants would prefer faculty-led seminars to discuss interactions with the simulated patients, while others saw greater potential in student-led discussions.

When asked about preferred timing for implementing the simulated patients in the curriculum, participants suggested three main

Table 2
Overview of the 5 main themes and 12 subthemes identified the thematic analysis.

Theme	Simulated but meaningful-bridging simulation and real-world	Variation matters: A wish for diverse patient representations	Feeling seen- balancing usefulness and overload in detailed feedback	Models vs faculty-structure, format and fairness of the feedback	Risks and opportunities of technological models: balancing efficiency and time-consuming challenges
Sub-theme	Usability and guidance for effective use	Realistic and engaging patient interactions	Relevance and structural quality of the feedback	Thoroughness and focus of the feedback	General risks and privacy aspects
Sub-theme	Pedagogical potential and preparation for future professional roles	Challenges with articulate and unrealistic expressions	Content validity and scope of the feedback	Objectivity and written format	Technological alternatives and user preferences
Sub-theme	Adaptability of the models and their use in education	The need for variation in patient profiles			

options: using them between role-play sessions to reinforce learning and provide additional practice before examinations; integrating them during practical pharmacy placements to gain more experience with patient cases and build confidence before real-life encounters; and incorporating them across multiple courses to support progression in learning. One participant expressed it as: *"I'd say during the internship, when you're in the process of learning and handling customer interactions at the same time to develop your skills. Also, based on my own experience during the internship, there weren't that many customers I could practice with, and I felt like I wanted more opportunities to interact with customers. I think it's better for both the customer and myself if I feel more confident in what I'm saying."* (Student 9).

Participants also felt that the chatbot tool had a positive impact on their preparation for future professional practice. Especially in a time when online-pharmacies are becoming increasingly common, they considered written communication training particularly valuable *"I think the model will primarily train students to use a chat in written communication, and I believe there will be a significant need for this among pharmacists in the future. Pharmacists will need to be able to write and communicate with patients in this way, just like this chatbot does."* (Faculty 5).

Variation matters: A wish for diverse patient representations

The majority of participants found the conversation with the virtual patients engaging and interesting. Many described it as 'fun' and 'enjoyable.' A common opinion was that the virtual patient felt realistic, particularly in how they expressed themselves and responded to different types of questions, which resembled the behavior of a real patient. The responses were perceived as being in everyday language. Participants also noted that the virtual patient's way of expressing concern enhanced the sense of realism, as it reflected emotions that real patients often convey. *"I thought she [the virtual patient] was very realistic and expressed feelings of concern—it really felt like she was worried."* (Student 9).

Many emphasized the importance of virtual patients representing a variety of communication styles, levels of engagement, and health literacy in order to make the training more realistic and better prepare students for interacting with diverse patients. Despite the positive aspects, some faculty and students noted that the virtual patient occasionally came across as unrealistic. A recurring observation was that it seemed overly knowledgeable and articulate, describing symptoms and asking follow-up questions in a way more typical of educational material than real-life patient interactions. *"I mean, you didn't have to pull the answers out of her like you sometimes have to do. I felt like you got quite a lot of information right from the start. That's probably not something all customers do."* (Student 1).

Participants also pointed out that the virtual patient always understood the questions correctly and provided exactly the information pharmacists were looking for, which reduced the sense of authenticity. This fluency also made the interaction feel more like a conversation at a physical pharmacy than the more concise and fragmented communication anticipated in online pharmacy chats. Nonetheless, most participants valued the clarity and knowledgeability of the virtual patient, noting that it helped lower the cognitive load during early training sessions. Some suggested a progression in difficulty; starting with articulate patient cases and gradually introducing more brief or complex scenarios to reflect more realistic variation.

Feeling seen: Balancing usefulness and overload in detailed feedback

Several participants appreciated that the automated feedback was clearly structured, divided into sections focusing on specific communication skills. The feedback was considered well designed, offering both detailed observations and concrete suggestions for improvement. Overall, participants felt encouraged by knowing what they did well and saw the feedback as a relevant and practical tool for improving their communication skills. *"I mean, it highlights both the positive and negative aspects, which is great. It feels like I get to know what I'm good at and what I can improve on, and that's really the whole point of feedback. That's why I think it's good."* (Student 3).

However, a few participants noted occasional inaccuracies in the automated feedback upon closer review. These could sometimes lead to confusion and a sense that the feedback was not entirely reliable. Despite this, most participants felt that the feedback was generally accurate and aligned with their own perception of areas needing improvement. *"It brought up things that I've heard before, but also new things that I hadn't thought about and that no one else has mentioned."* (Student 4). Some participants felt the feedback was too long and occasionally repetitive. Others valued the detail, as it helped them understand both their strengths and areas for improvement. The summary at the end (Appendix C) was particularly valued, as it provided an overview of the conversation, especially useful when reviewing multiple feedback sessions. Participants also expressed a desire for the automated feedback to cover a broader range of communication skills. For example, they suggested including insights into the communicator's overall approach, how incorrect word order and technical terms might lead to misunderstandings and identifying possible signs of bias that could negatively affect the quality of advice. Additionally, both students and faculty pointed out that feedback on pharmacotherapy should be strengthened, as this content is essential for evaluating whether appropriate clinical advice has been given.

Another aspect raised was the need for progressive feedback, where each session builds upon the previous one and acknowledges areas where the student has already improved. This was seen as a way to reinforce confidence by not always focusing solely on what needs to change. One participant reflected on the risk of constantly receiving suggestions for improvement—even when unnecessary—as this could lead to insecurity and overanalyzing one's performance. *"In some cases, if you're confident in how you phrase something, but you're constantly told there's something that needs improvement, you might start to question yourself a bit."* (Student 5).

AI vs faculty: Structure, format and fairness of the feedback

Students generally felt that the AI-generated feedback was better than what they typically receive from faculty in similar training

situations based on the content of the feedback and the balance between positive and constructive remarks. Faculty also noted a positive difference compared to the feedback they usually provide themselves. *"I also think that it [the system] analyzes everything much more thoroughly and probably doesn't miss things in the same way we do. For us, it's about remembering what we remember, and we might not recall everything we heard or saw. Instead, we might only remember the most important things that stood out to us."* (Faculty 2). Students especially appreciated that the AI-generated feedback was more detailed and placed stronger emphasis on communication skills—an area they felt is often overlooked in traditional faculty feedback in self-care consultation teaching.

Several participants mentioned that limited time often prevents faculty from giving thorough, individualized feedback, particularly on communication. In contrast, the automated feedback allowed students to review the information at their own pace and reflect more deeply: *"I find it difficult to take in information and remember things that are said verbally, so I think this is a much calmer way to practice communication. And above all, you can review it afterward."* (Student 6). The automated feedback was also perceived as more standardized and fairer. The fact that all students were assessed by the same system contributed to a sense of equality and consistency in evaluation.

Risks and opportunities of technological models: Balancing efficiency and time-consuming challenges

Regarding potential security risks, all participants perceived the use of the chatbot as safe, noting that the scenarios are simulated and no real patient data or personal information is used or shared during the interaction. Some participants expressed concerns that chat-based interactions may limit the development of essential non-verbal communication skills, which are particularly important in face-to-face pharmacy settings. *"The thing about body language and non-verbal cues that you can't see here is that, when you later go out into practice, you might not know how to apply those skills when you're interacting with real customers."* (Student 5).

Faculty members pointed out that conversations with the simulated patients were time-consuming. In comparison, spoken interactions were perceived as more time-efficient, as they do not require the same effort in formulating written chat responses. There was concern that the time demands of chat-based tasks could reduce students' capacity to focus on other learning activities. Many participants saw strong potential in developing voice-based functionality, suggesting that a microphone-enabled version could provide a more realistic learning experience. Capturing voice and tone was seen as particularly valuable in helping students assess a virtual patient's emotional state and better understand the context of their condition. At the same time, several participants found the current chatbot tool fully adequate for the intended training purposes. They highlighted the benefits of being able to type responses at their own pace and reflect more deeply by reviewing written feedback afterward. A commonly suggested improvement was to offer a combination of both chat- and voice-based interactions.

Analysis and discussion of educational activity

This study explored the potential of LLM-based chatbots to support pharmacy students in developing written communication skills relevant to self-care consultations. As online pharmacies continue to expand globally, the need for effective training in written communication becomes increasingly apparent. To our knowledge, this is the first study exploring pharmacy students' and faculty experiences with simulated patients and LLM-based feedback, providing insights for future AI-based tools in pharmacy education.^{16,17} Three key insights emerged: First, students perceived the simulated patient cases as realistic, highlighting the potential of these tools to support authentic learning experiences and suggesting opportunities for future co-development of scenarios and testing of simulated patient interactions. Secondly, both student and faculty appreciated the structure, detail and fairness of AI-generated feedback. Students valued its focus on communication skills, an area they felt is often overlooked in traditional faculty feedback, while faculty noted the thoroughness and consistency of the evaluation. Lastly, students recognised the cognitive and emotional dimensions of these interactions, emphasising the need for further refinement to balance cognitive load and emotional realism.

Authenticity and co-development of simulated patients

One of the most striking findings was students' ability to perceive authenticity of the simulated patient interactions. Most participants described the simulated patients as realistic, particularly in their emotional expression.²⁸ Moreover, the ability of simulated patients to mimic real-life scenarios has been shown to enhance learner engagement and skill acquisition.²⁹ Our results show that students can be involved in creating cases with simulated LLM models, something rarely seen before, where cases have been developed by experts or researchers using real cases, despite its clear potential.⁹ The importance of involving learners in the design process is supported by evidence that well-designed interventions improve learning outcomes in virtual patient-based training.^{28,30} To improve realism, diversity and robustness, future development could benefit from establishing a shared environment of virtual patients co-developed with pharmacists, students, and other stakeholders. Such collaborative open infrastructure would enable the pooling of expertise and use cases for more authentic and complex scenarios.

The evolving role of educator feedback in an AI-driven learning environment

The fact that both students and faculty rated AI-generated feedback as superior to faculty feedback should not only be seen as a strength of AI, but as a signal that faculty feedback practices need to be improved. Students valued the AI's structured, detailed, and impartial responses, qualities that are sometimes lacking in traditional feedback due to time constraints, variation in faculty approaches, or insufficient training in providing effective feedback. This highlights a need for faculty development, not only to better

leverage AI as a pedagogical tool, but also to ensure that human feedback remains pedagogically meaningful, consistent, and valued. Rather than being replaced, educators must embrace a new role. A one that integrates AI as a complement while reinforcing the uniquely human aspects of facilitation, reflection, and judgement. A summary of the assessment tools reviewed by McLaughlin et al. could be valuable to include in further work on feedback, especially if AI is to be used as a complementary method in assessments and to prepare students for such integration.³¹

Cognitive and emotional dimensions of AI-learning

Students noted that chatbot interactions could be tailored to match their training level and cognitive capacity. In early communication training, the structured and clear dialogues can help reduce cognitive load, allowing students to focus on mastering basic communication skills. As students progress, these interactions could evolve to include more complex cases. This approach aligns with Cognitive Load Theory, which emphasizes the importance of managing cognitive demands to optimize learning.^{32,33} At the same time, students identified limitations in emotional realism. While the chatbot successfully conveyed basic emotional expressions, the lack of non-verbal cues, such as tone, facial expressions, and body language, was identified as a drawback, particularly when preparing for emotionally complex, in-person interactions. However, this limitation is less relevant when the goal is to train students for online or written pharmacy communication, where such cues are naturally absent. Participants expressed interest in more advanced voice-based or hybrid models that incorporate both verbal and non-verbal elements. Although such technologies exist, they are still under development, often less robust, and relatively expensive.¹⁵

Implications for faculty

While this study used OpenAI's LLM, the approach is not dependent on a specific provider; rather, its effectiveness depends on well-crafted prompts (see Appendix D as one example) grounded in a clear theoretical framework. Therefore, it is suggested to incorporate both communication theories as well as pedagogical frameworks including assessment tools³¹ into the development process of prompts. Readers may use our prompt as a starting point (see Appendix D) and also refer to Section 2.1 (Study Design) for details on how we designed the simulated patients (e.g. Table 1 and Appendix D) and suggestions from the Evaluation Findings section on how to improve them and the associated feedback process (Appendix D). For instance, we applied a consultation framework widely used in pharmacy education and in community pharmacies in Sweden, known as The Communication Ladder (Appendix B). This framework, however, can be substituted with other relevant feedback models. The communication skills outlined in Appendix D reflect how we structure our teaching, but they can readily be adapted to different educational or professional contexts.

In our project, students were provided access to a paid ChatGPT account. For broader implementation, we recommend collaborating with an IT developer to establish a dedicated platform, based on ChatGPT or another LLM, where students do not require individual subscriptions. Such an arrangement is particularly important if the activity is to be implemented at scale, as it ensures a neutral, institutionally managed platform for student use. Importantly, the platform does not need to be technically sophisticated; rather, it should prioritize accessibility, reliability, and ease of use. Ensuring that these tools are freely accessible to students is critical for promoting equitable education. Universities should prioritize the development of cost-effective and environmentally sustainable models that do not impose financial burdens on students.

Additionally, initiating and supporting an international collaboration among pharmacy educators to co-develop a shared virtual patient infrastructure could significantly enhance the realism, diversity, and authenticity of case-based learning. Faculty must also consider how to best leverage their role to support student learning when teaching with simulated patients, a topic needing further study. It is also crucial to provide clear instructions about what students can expect from the learning activity, for example, by explaining the context of the simulated patient, how the chatting works, and discussing potential limitations of the model and how these affect learning. The development in ChatGPT was convenient and straightforward; most of our time was spent on building the framework not on technical details. The provided prompt is model agnostic and should work in any LLM of choice. However, as participants highlighted, the model can produce hallucinations, and it is important to address this in teaching. It is also necessary to consider reviewing the prompts in other or newer models if they have not been used for some time to ensure proper functionality. The rise of online pharmacies underscores the need for empathic written communication skills,^{18,19,34} in chat-based consultation, AI-based tools may support this learning.

Strengths and limitations

A key strength of this study is the development of an AI chatbot tool that combines patient simulation with detailed feedback. Participants valued the structured feedback, which included both positive reinforcement and constructive suggestions for improvement. This study offers an early-stage evaluation of AI-supported communication training, enabling systematic identification of pedagogical and technical considerations before large-scale implementation, it also highlights the need for further research. For example, future studies should evaluate the impact of these models on students' communication skills, thereby addressing a gap identified in previous reviews.^{9,12} The inclusion of both students and faculty provided diverse perspectives. Saturation was considered achieved when no new codes or meaningful variations emerged from the final interviews, indicating that the data adequately captured the range of perspectives relevant to the study aims. However, the study was conducted at a single university with a predominance of female participants, which may limit transferability. Participants only tested one chatbot each. Future studies should explore broader implementation contexts, include more diverse user groups, and examine long-term learning outcomes. The automated feedback

occasionally contained inaccuracies, which could undermine its reliability. While these instances were relatively rare, they highlight the need for ongoing refinement of the system.

Summary of analysis

This study demonstrates the potential use of AI chatbots to enhance pharmacy students' written communication skills in self-care consultations. By addressing the limitations and building on the strengths identified in this research, AI tools can play an innovative role in pharmacy education, preparing students for the challenges of modern pharmacy practice.

During the preparation of this work the author(s) used ChatGPT 4.0 in order to improve the language and reference list. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.cptl.2025.102503>.

References

- Hovland R, Bremer S, Frigaard C, et al. Effect of a pharmacist-led intervention on adherence among patients with a first-time prescription for a cardiovascular medicine: a randomized controlled trial in Norwegian pharmacies. *Int J Pharm Pract*. 2020;28(4):337–345. <https://doi.org/10.1111/ijpp.12598>.
- Street Jr RL, Makoul G, Arora NK, Epstein RM. How does communication heal? Pathways linking clinician-patient communication to health outcomes. *Patient Educ Couns*. 2009;74(3):295–301. <https://doi.org/10.1016/j.pec.2008.11.015>.
- Kaae S, Rossing C, Husted GR, Fosgerau CF. How patient-centredness takes place in pharmacy encounters: a critical common-sense interpretation of video-recorded meetings. *Int J Clin Pharmacol*. 2023;45(1):146–153. <https://doi.org/10.1007/s11096-022-01508-2>.
- Greenhill N, Anderson C, Avery A, Pilnick A. Analysis of pharmacist-patient communication using the Calgary-Cambridge guide. *Patient Educ Couns*. 2011;83(3):423–431. <https://doi.org/10.1016/j.pec.2011.04.036>.
- Wallman A, Vaudan C, Sporrang SK. Communications training in pharmacy education, 1995–2010. *Am J Pharm Educ*. 2013;77(2):36. <https://doi.org/10.5688/ajpe77236>.
- Kurtz S, Draper J, Silverman J. *Teaching and Learning Communication Skills in Medicine*. 2nd ed. Radcliffe Publishing; 2005.
- Svensberg K, Sporrang SK, Lupattelli A, Olsson E, Wallman A, Björnsdóttir I. Nordic pharmacy students' opinions of their patient communication skills training. *Am J Pharm Educ*. 2018;82(2):6208. <https://doi.org/10.5688/ajpe6208>.
- Svensberg K, Björnsdóttir I, Wallman A, Sporrang SK. Nordic pharmacy schools' experience in communication skills training. *Am J Pharm Educ*. 2017;81(9):6005. <https://doi.org/10.5688/ajpe6005>.
- Bowers P, Graydon K, Ryan T, Lau JH, Tomlin D. Artificial intelligence-driven virtual patients for communication skill development in healthcare students: a scoping review. *Australas J Educ Technol*. 2024;40(3):39–57. <https://doi.org/10.14742/ajet.9307>.
- Butow P, Hoque E. Using artificial intelligence to analyse and teach communication in healthcare. *Breast*. 2020;50:49–55. <https://doi.org/10.1016/j.breast.2020.01.008>.
- Amirthalingam P, Pakkiri Mohamed SH, Veeramani VP, et al. Comparison of MyDispense with in-person simulation in drug information training of pharmacy curriculum: a randomized cross-over study. *BMC Med Educ*. 2025;25(1):356. <https://doi.org/10.1186/s12909-025-06921-5>.
- Stamer T, Steinhäuser J, Flägel K. Artificial intelligence supporting the training of communication skills in the education of health care professions: scoping review. *J Med Internet Res*. 2023;25, e43311. <https://doi.org/10.2196/43311>.
- Liaw SY, Tan JZ, Lim S, et al. Artificial intelligence in virtual reality simulation for interprofessional communication training: mixed-method study. *Nurse Educ Today*. 2023;122, 105718. <https://doi.org/10.1016/j.nedt.2023.105718>.
- Hershberger PJ, Pei Y, Bricker DA, et al. Motivational interviewing skills practice enhanced with artificial intelligence: ReadMI. *BMC Med Educ*. 2024;24(1):237. <https://doi.org/10.1186/s12909-024-05217-4>.
- Barker LA, Moore JD, Cook HA. Generative artificial intelligence as a tool for teaching communication in nutrition and dietetics education—a novel education innovation. *Nutrients*. 2024;16(7):914. <https://doi.org/10.3390/nu16070914>.
- Abdel Aziz MH, Rowe C, Southwood R, et al. A scoping review of artificial intelligence within pharmacy education. *Am J Pharm Educ*. 2024;88(1), 100615. <https://doi.org/10.1016/j.ajpe.2023.100615>.
- Nakagawa N, Odanaka K, Ohara H, Kisara S. Communication training for pharmacy students with standard patients using artificial intelligence. *Curr Pharm Teach Learn*. 2022;14(7):854–862. <https://doi.org/10.1016/j.cptl.2022.06.021>.
- Almaman A. The digital transformation in pharmacy: embracing online platforms and the cosmeceutical paradigm shift. *J Health Popul Nutr*. 2024;43(1):60. <https://doi.org/10.1186/s41043-024-00550-2>.
- Pjenggaard D, Rasmussen LL, Rossing CV, Nørgaard LS. *Pros and cons of counselling at online-only pharmacies: a mixed-methods case study*. Preprint posted May. 2024. <https://doi.org/10.21203/rs.3.rs-4354170/v1>.
- Dave T, Athaluri SA, Singh S. ChatGPT in medicine: an overview of its applications, advantages, limitations, future prospects, and ethical considerations. *Front Artif Intell*. 2023;6, 1169595. <https://doi.org/10.3389/frai.2023.1169595>.
- Beardsley RS, Kimberlin CL, Tindall WN. *Communication Skills in Pharmacy Practice: A Practical Guide for Students and Practitioners*. 7th ed. Wolters Kluwer Health; 2019.

22. Ayad S, Alsayoud F. Prompt engineering techniques for semantic enhancement in business process models. *Bus Process Manag J.* 2024;30(7):2611–2641. <https://doi.org/10.1108/BPMJ-02-2024-0108>.
23. Kallio H, Pietilä AM, Johnson M, Kangasniemi M. Systematic methodological review: developing a framework for a qualitative semi-structured interview guide. *J Adv Nurs.* 2016;72(12):2954–2965. <https://doi.org/10.1111/jan.13031>.
24. Kirkpatrick JD, Kirkpatrick WK. *Kirkpatrick's Four Levels of Training Evaluation*. ATD Press; 2016.
25. Graneheim UH, Lundman B. Qualitative content analysis in nursing research: concepts, procedures and measures to achieve trustworthiness. *Nurse Educ Today.* 2004;24(2):105–112. <https://doi.org/10.1016/j.nedt.2003.10.001>.
26. Tjora A. *Kvalitative Forskningsmetoder I Praksis [Qualitative Research Methods in Practice]*. 4th ed. Gyldendal; 2021.
27. Swedish Ethical Review Authority. What the Act Says. <https://etikprovningsmyndigheten.se/en/what-the-act-says>; 2025. Accessed June 16, 2025.
28. Lee J, Kim H, Kim KH, Jung D, Jowsey T, Webster CS. Effective virtual patient simulators for medical communication training: a systematic review. *Med Educ.* 2020;54(9):786–795. <https://doi.org/10.1111/medu.14152>.
29. Musa D, Gonzalez L, Penney H, Daher S. Technology acceptance and authenticity in interactive simulation: experimental study. *JMIR Med Educ.* 2023;9, e40040. <https://doi.org/10.2196/40040>.
30. Könings KD, Mordang S, Smeenk F, Stassen L, Ramani S. Learner involvement in the co-creation of teaching and learning: AMEE guide no. 138. *Med Teach.* 2021; 43(8):924–936. <https://doi.org/10.1080/0142159X.2020.1838464>.
31. McLaughlin JE, Kelley K, Mortha SM, Bowen JF. Tools for assessing communication in pharmacy education: review and recommendations. *Am J Pharm Educ.* 2024;88(12), 101328. <https://doi.org/10.1016/j.ajpe.2024.101328>.
32. van Merriënboer JJJ, Sweller J. Cognitive load theory in health professional education: design principles and strategies. *Med Educ.* 2010;44(1):85–93. <https://doi.org/10.1111/j.1365-2923.2009.03498.x>.
33. Leppink J, van den Heuvel A. The evolution of cognitive load theory and its application to medical education. *Perspect Med Educ.* 2015;4(3):119–127. <https://doi.org/10.1007/s40037-015-0192-x>.
34. Rollnick S, Miller WR, Butler CC. *Motivational Interviewing in Health Care: Helping Patients Change Behavior*. 2nd ed. Guilford Press; 2022. ISBN 9781462550371.